HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2023)

Student No.: 02 Name: _____ Marks: / 40

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. Let $h(x) = ax^6 + 16x^3 + b$, where a and b are constants. If $h(x)$ is divisible by $2x - 3$, find the remainder when $h(x)$ is divided by $2x + 3$.

   A. -108

   B. -54

   C. 54

   D. 108

(DSE-2023 PII Q09)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

2. Which of the following statements about the graph of $y = 5 + (x-3)^2$ is true?

A. The graph opens downwards.

B. The x-intercept of the graph is 3.

C. The y-intercept of the graph is 5.

D. The graph passes through the point $(3,5)$.

(DSE-2023 PII Q10)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

3. The scale of a map is $1 : 50000$. If the actual area of an airport is $10\,km^2$, then the area of this airport on the map is

A. $2\,cm^2$.

B. $4\,cm^2$.

C. $20\,cm^2$.

D. $40\,cm^2$.

(DSE-2023 PII Q12)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

4. Let a_n be the nth term of a sequence. If $a_6 = 23$, $a_8 = 60$ and $a_{n+2} = a_{n+1} + a_n$ for any positive integer n, then $a_3 =$

A. 4.

B. 5.

C. 9.

D. 14.

(DSE-2023 PII Q14)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

5. The length of a side of a solid cube is $60\,\text{cm}$. The volume of a solid right circular cylinder is equal to the volume of the cube while the curved surface area of the circular cylinder is equal to the total surface area of the cube. Find the base radius of the circular cylinder.

A. $20\,\text{cm}$

B. $30\,\text{cm}$

C. $76\,\text{cm}$

D. $172\,\text{cm}$

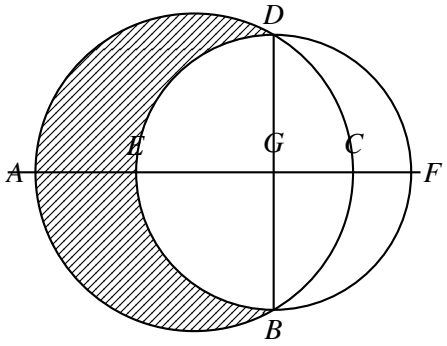
(DSE-2023 PII Q15)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

6. In the figure, AC is a diameter of the circle $ABCD$ while BD and EF are diameters of the circle $BEDF$. It is given that C and E lie on AF. Let G be the point of intersection of AF and BD. If $AG = 30\,\text{cm}$ and $CG = 10\,\text{cm}$, find the area of the shaded region correct to the nearest cm^2.



A. $209\,\text{cm}^2$

B. $367\,\text{cm}^2$

C. $383\,\text{cm}^2$

D. $540\,\text{cm}^2$

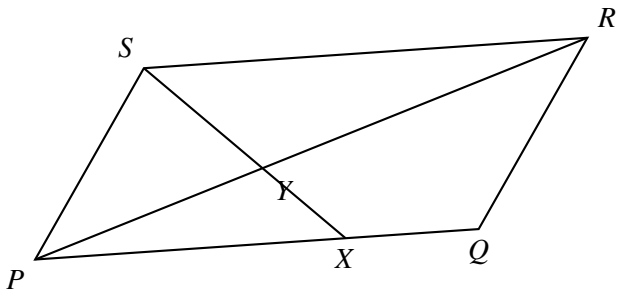
(DSE-2023 PII Q16)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

7. In the figure, $PQRS$ is a parallelogram. Let X be a point lying on PQ. Denote the point of intersection of PR and SX by Y. If the area of $\triangle PXY$ and the area of the quadrilateral $QRYX$ are $32\,\text{cm}^2$ and $58\,\text{cm}^2$ respectively, then the area of $\triangle RSY$ is



A. $40\,\text{cm}^2$.

B. $50\,\text{cm}^2$.

C. $58\,\text{cm}^2$.

D. $72\,\text{cm}^2$.

(DSE-2023 PII Q17)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

8. It is given that $ABCD$ is a rhombus. Denote the point of intersection of AC and BD by E. Which of the following must be true?

I. $AE = BE$

II. $\frac{AE}{AC} = \frac{BE}{BD}$

III. $AE^2 + BE^2 = CD^2$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

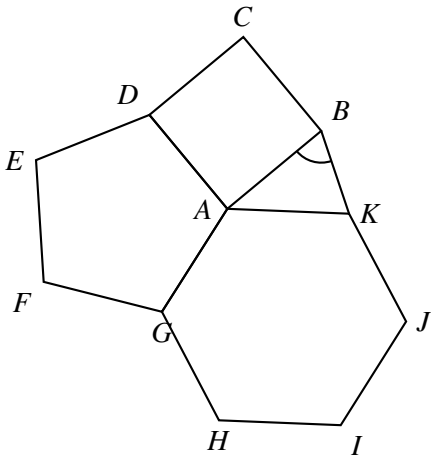
(DSE-2023 PII Q19)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

9. The figure shows the square $ABCD$, the regular pentagon $ADEFG$ and the regular hexagon $AGHIJK$. Find $\angle ABK$.



A. 69°

B. 72°

C. 74°

D. 75°

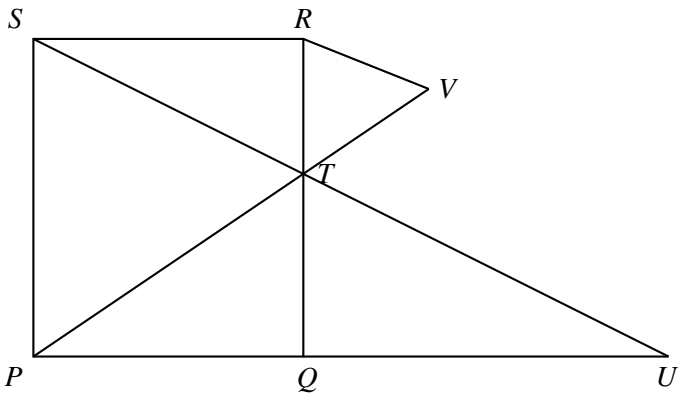
(DSE-2023 PII Q20)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

10. In the figure, $PQRS$ is a rectangle. Let T be a point lying on QR such that $\angle PTS = 90°$. PQ produced and ST produced meet at the point U. PT is produced to the point V such that $RT = RV$. Which of the following must be true?



A. $RV \parallel ST$

B. $\angle PTQ = \angle RTS$

C. $\triangle PST \sim \triangle UTQ$

D. $\triangle PQT \cong \triangle TRS$

(DSE-2023 PII Q21)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

11. The rectangular coordinates of the point P are $(\sqrt{2}, -\sqrt{2})$. If P is rotated anticlockwise about the origin through $90°$, then the polar coordinates of its image are

A. $(\sqrt{2}, 45°)$.

B. $(\sqrt{2}, 225°)$.

C. $(2, 45°)$.

D. $(2, 225°)$.

(DSE-2023 PII Q24)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

12. $1011001011001011_2 =$

A. $11 \times 2^{11} + 11 \times 2^5 + 11$.

B. $11 \times 2^{12} + 11 \times 2^6 + 11$.

C. $11 \times 2^{13} + 11 \times 2^7 + 11$.

D. $11 \times 2^{14} + 11 \times 2^8 + 11$.

(DSE-2023 PII Q31)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

13. Let $f(x) = 3x^2 + 18mx + 22m^2$, where m is a real constant. Which of the following statements about the graph of $y = -f(3x)$ must be true?

I. The x-coordinate of the vertex of the graph is m.

II. The y-coordinate of the vertex of the graph is $5m^2$.

III. The equation of the axis of symmetry of the graph is $x + m = 0$.

A. I only

B. II only

C. I and III only

D. II and III only

(DSE-2023 PII Q35)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

14. Let $T(n)$ be the nth term of an arithmetic sequence. If $T(11) = 83$ and $T(25) + T(30) = 463$, find the least value of k such that $T(1) + T(2) + T(3) + \cdots + T(k) > 4 \times 10^5$.

A. 299

B. 300

C. 944

D. 945

(DSE-2023 PII Q36)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

15. Consider the following system of inequalities:

$$\begin{cases} x+3 \geq 0 \\ 2x+3y-12 \leq 0 \\ 5x-3y+12 \leq 0 \end{cases}$$

Let D be the region which represents the solution of the above system of inequalities. Find the range of values of β such that the greatest value of $\beta x+6y$ is 24, where (x, y) is a point lying in D.

A. $\beta \leq -10$

B. $\beta \geq -10$

C. $\beta \leq 4$

D. $\beta \geq 4$

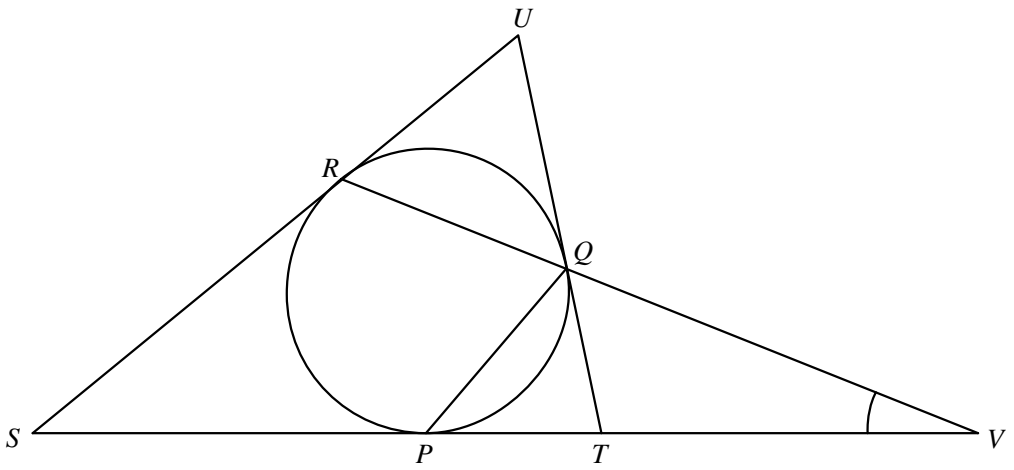
(DSE-2023 PII Q37)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

16. In the figure, P, Q and R are points lying on a circle. ST, TU and SU are the tangents to the circle at P, Q and R respectively. RQ produced and ST produced meet at the point V. If $\angle PSR = 34^\circ$ and $\angle QPT = 46^\circ$, then $\angle PVQ =$



A. 17°.

B. 22°.

C. 27°.

D. 28°.

(DSE-2023 PII Q38)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

17. The straight line $hx + ky = 6$ and the circle $x^2 + y^2 - 8x - 4y - 18 = 0$ intersect at the points M and N, where h and k are constants. If the coordinates of the mid-point of MN are $(1, 0)$, find k.

A. 4

B. 6

C. 9

D. 12

(DSE-2023 PII Q39)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

18. The base of the right pyramid $VABCD$ is the square $ABCD$. Let θ be the angle between $\triangle ABV$ and $\triangle BCV$. If $AB : AV = 5 : 4$, then $\cos\theta =$

A. $\dfrac{-25}{39}$.

B. $\dfrac{-17}{33}$.

C. $\dfrac{-9}{16}$.

D. 0.

(DSE-2023 PII Q40)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

19. There are 15 teachers in a group. If 5 teachers are selected from the group to form a committee consisting of 1 chairperson and 4 members, how many different committees can be formed?

A. 3003

B. 15015

C. 20475

D. 360360

(DSE-2023 PII Q42)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

20. When a boy throws a dart, the probability that he hits the target is 0.6 . If this boy throws the dart 4 times, find the probability that he hits the target at least 2 times.

A. 0.5248

B. 0.7056

C. 0.8208

D. 0.8464

(DSE-2023 PII Q43)

I can finish _____ questions

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.